Computer Networks

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- **What ?**
- ✔ A group or system of interconnected people or things
- **Why?**
- ✔ Connection
- ✔ Communication
- **Where?**
- ✔ When we need to transfer anything.

Computer Networks

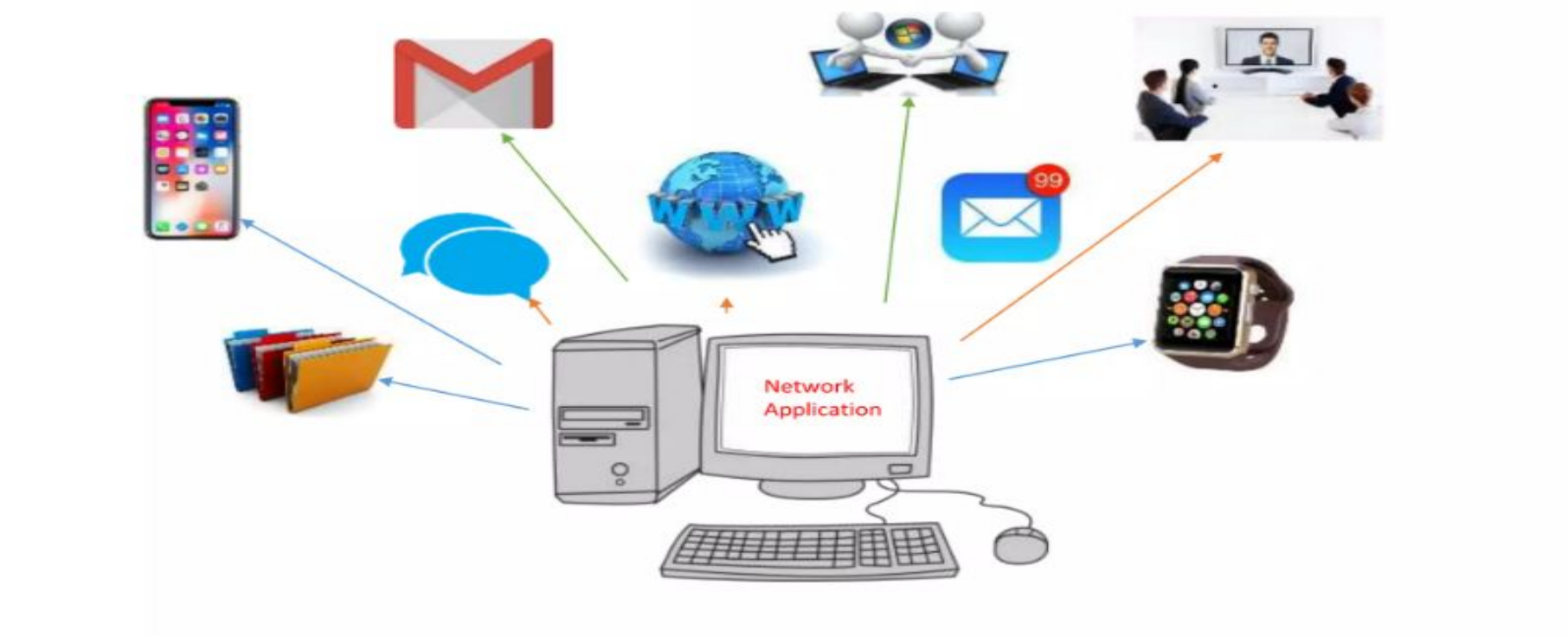
- A computer network is a set of devices connected by communication links.

- A node can be a computer, printer or any other device capable of sending or receiving data generated by other nodes on the network.

Computer Network

Networking

- Networking is a process of communication between two or more remote parties, that involves the connection of computers media and networking devices.



Advantages of Networking

- **Easy Communication**

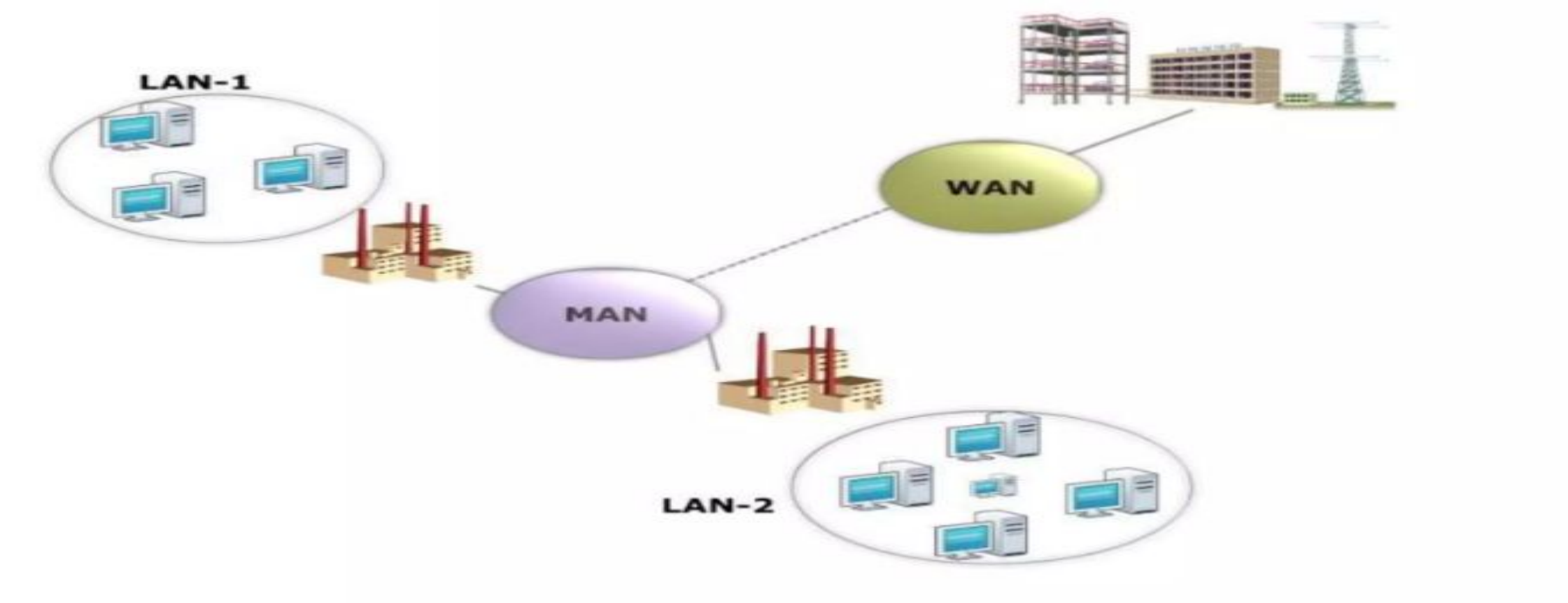
- File, data and information sharing
- Resource sharing(hardware)
- Increase storage capacity
- Reduce cost
- Save time

Network Components:

- **Devices:** Computers, servers, routers, switches, and other hardware connected to the network.
- **Cables and Connections:** Ethernet cables, fiber optic cables, and wireless connections (Wi-Fi).
- **Protocols:** Sets of rules governing communication between devices, such as TCP/IP, HTTP, and DNS.

Types of Network

- Physical structure-LAN-WAN-MAN



LAN (Local Area Network): A network confined to a small geographic area, such as a home, school, or office.

WAN (Wide Area Network): A network that spans a large geographic area, often connecting multiple LANs across cities or countries.

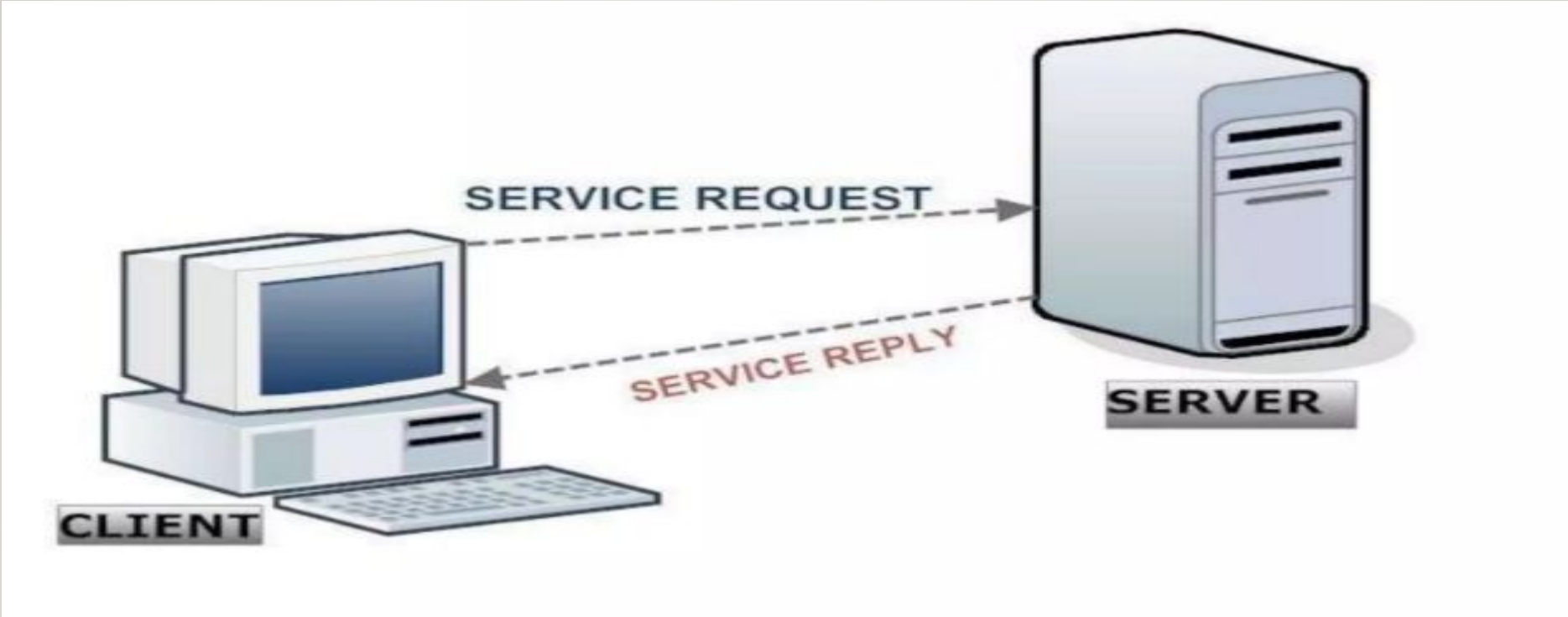
Metropolitan Area Network (MAN): A MAN is a network that spans a city or metropolitan area, providing connectivity between LANs within the same region.

Network Terms

- Host- Workstation- Server- Client-Node

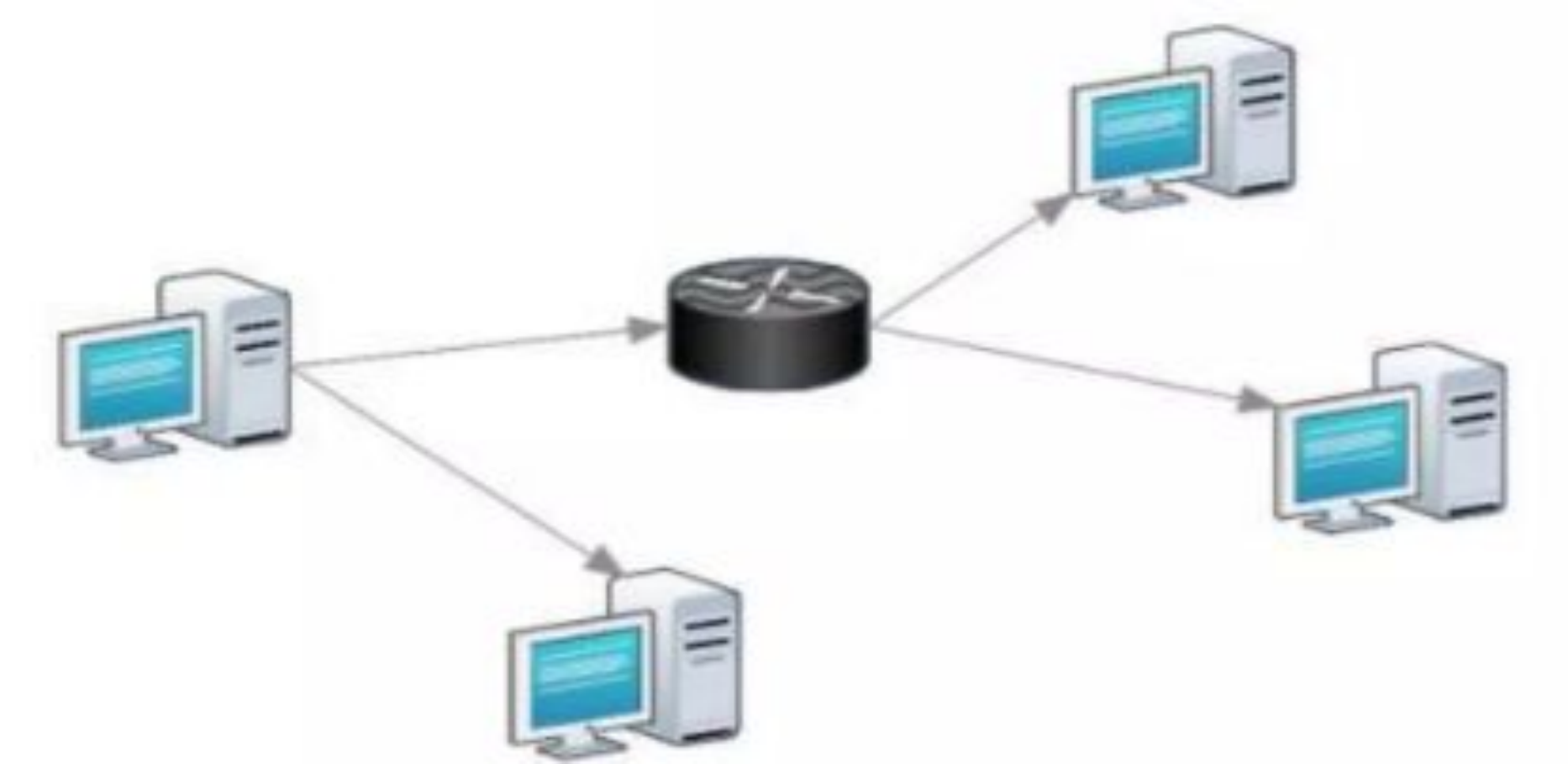
- **Host:** A network host is a computer or other device connected to a computer network.
- **Workstation:** A workstation is a computer dedicated to a user or group of users engaged in business or professional.

Client and Server



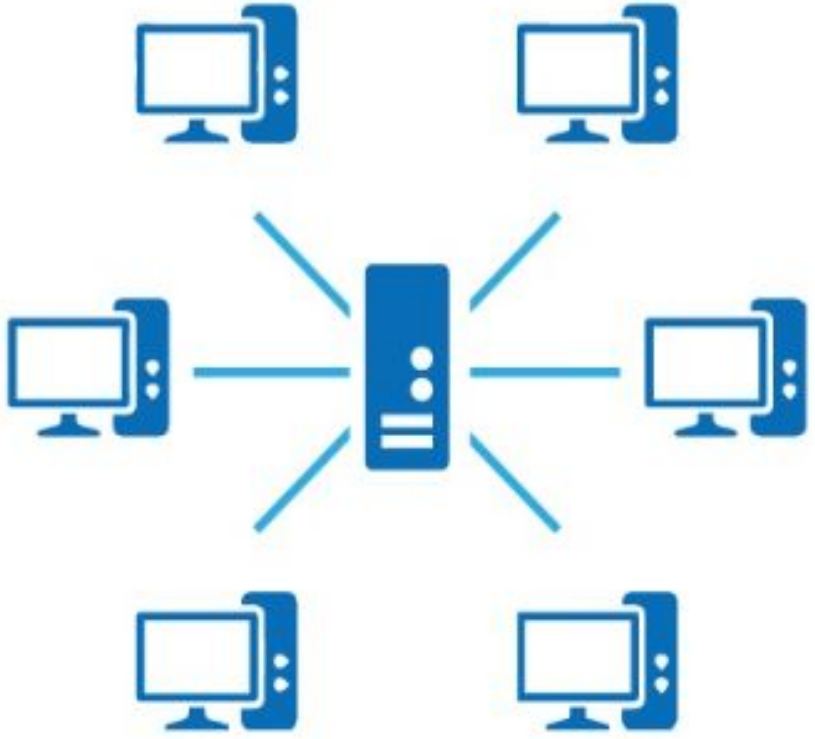
- For process to process communication we need Client Server Paradigm.
- A process om local host, called a client, needs Services from a process usually on the remote host called a server.
- A server is an a computer program that accepts and responds to requests made by another program, known as client.

- **Node:** A node can be a computer or some other device such as printer that can receive, create, store or send data along distributed network routes.

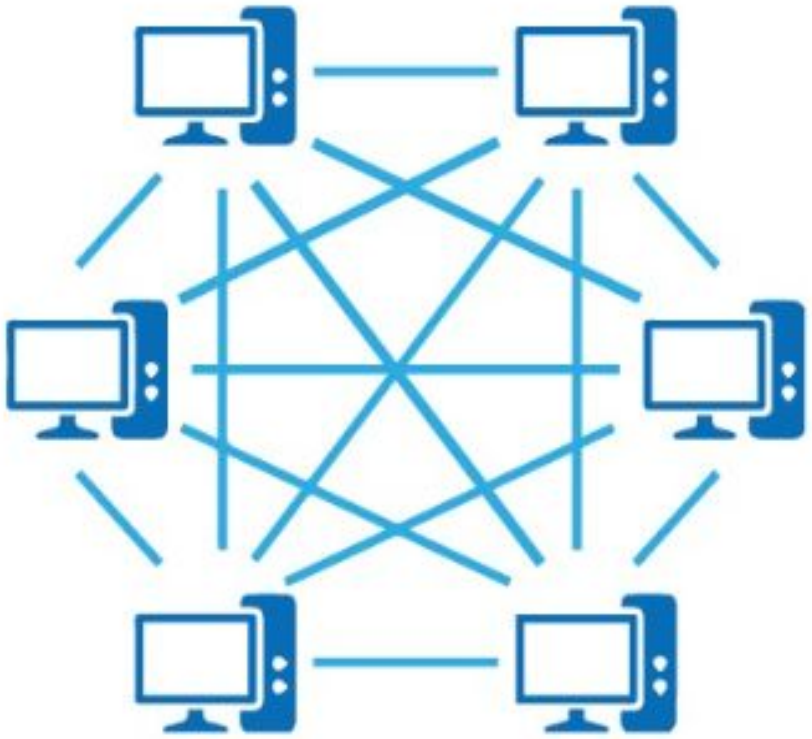


Types Of Network Architecture

- **Peer to Peer:** A peer to peer network is one in which two or more PCs share files and access to devices such as printers without requiring a separate server computer or server software.
- **Client-Server Network:** In client-server model, where client request for a service to a service and server replies according to client request.



A **Server** based Network

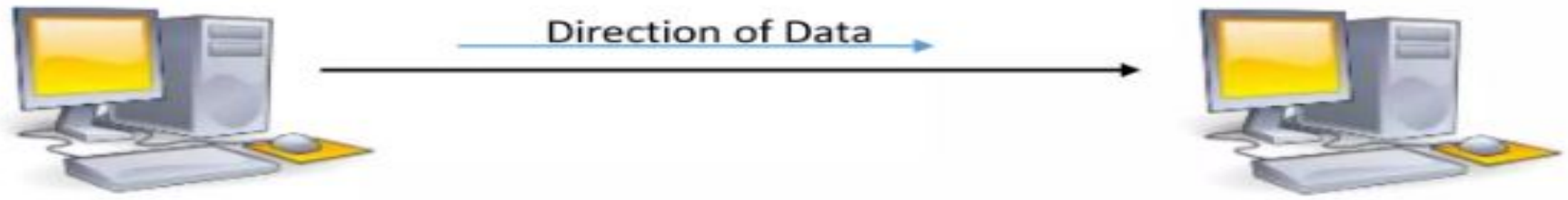


A **Peer-to-Peer** based Network

Data Flow

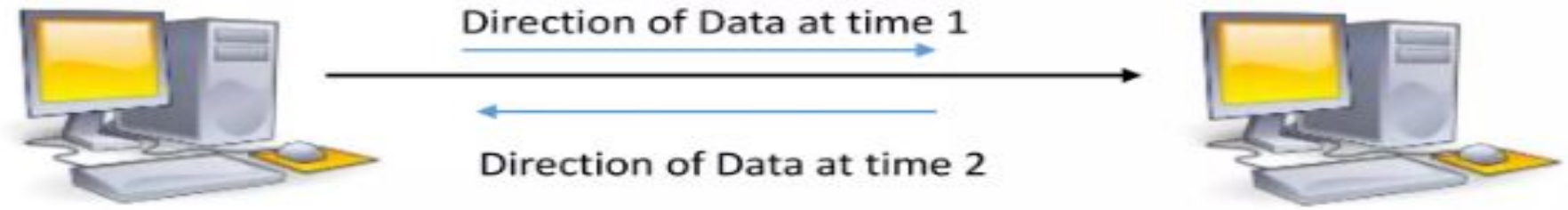
- Simplex- Half Duplex- Full Duplex

- In Simplex mode:
  - Communication is unidirectional,
  - Only one can transmit and other can receive.
  - Ex: keyboard and monitor



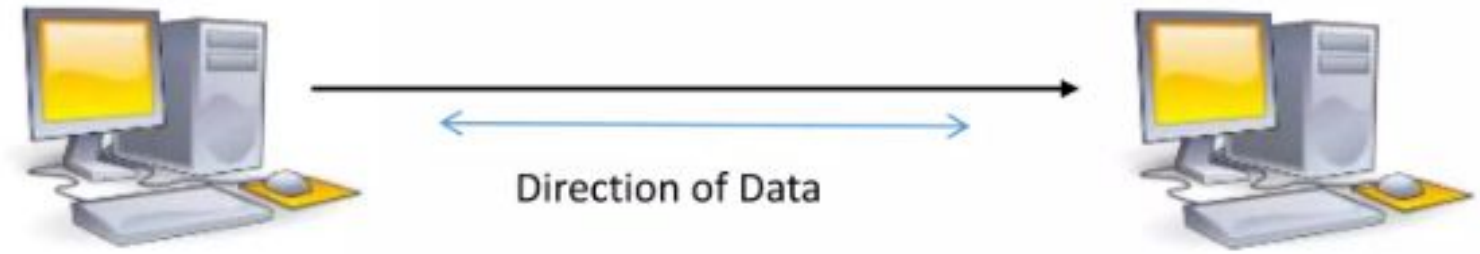
Simplex Communication Structure

- In Half Duplex mode:
  - Both can transmit and receive,
  - But not at a same time.
  - At a time, only one can send or receive.
  - Ex: Walkie-talkies



Half Duplex Communication Structure

- In Full Duplex mode:
  - Both station can transmit and receive simultaneously,
  - But at a same time.
  - Ex: Telephone Networks



Full Duplex Communication Structure

Workgroup Vs Domain

- **Workgroup:**

- A **workgroup** is a peer-to-peer network using common software. A workgroup allows all participating and connected systems to access shared resources such as files, system resources and printers.

Domain:

- In a **domain** is a subnetwork made up of a group of clients and servers under the control of one central security database.

Workgroup Vs Domain

Domain	Workgroup
• Computers on workplace networks are usually part of a domain.	• Computers on home networks are usually part of a workgroup.
• One or more computers are server.	• All computers are peers.
• For data security.	• No data security.